Chapter 8: File Handling in C

File handling is a crucial aspect of programming that enables you to store, retrieve, and manipulate data in external storage devices like hard drives or cloud servers. In C programming, file handling allows you to interact with files, perform operations like reading from and writing to files, and manage file input and output (I/O) effectively. This chapter will explore how to work with files in C, focusing on opening, reading, and writing files, the various file modes and operations, and how to handle errors that may occur during file processing.

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1. Opening, Reading, and Writing Files

1.1 Opening Files

Before you can perform any operation on a file, it must be opened. In C, the `fopen()` function is used to open a file. The syntax for the `fopen()` function is:

```
FILE *fopen(const char *filename, const char *mode);
```

Where:

- `filename` is the name of the file you want to open.
- `mode` specifies the operations that can be performed on the file (read, write, append, etc.).